

## **Advanced Java Programming**

**Course Title:** Advanced Java Programming

**Course No:** CSC409

**Nature of the Course:** Theory + Lab

**Semester:** VII

**Full Marks:** 60 + 20 + 20

**Pass Marks:** 24 + 8 + 8

**Credit Hrs:** 3

### **Course Description:**

This course familiarizes students with basic as well as advanced features of Java Programming. Emphasis will be given to GUI and event-driven programming, Database Connectivity, Socket Programming, Servlets and JSP Technology, and Distributed Programming.

### **Course Objectives:**

The main objective of this course is to

- Introduce basic concepts of Java Programming.
- Exemplify the concept of GUI programming and JDBC
- Demonstrate socket programming, remote objects, and servlet and JSP Technology

### **Course Contents:**

#### **Unit 1: Programming in Java (8 Hrs.)**

- 1.1. Java Architecture, Java Buzzwords, Path and ClassPath variables, Sample Java Program, Compiling and Running Java Programs.
- 1.2. Arrays, for each loop, Class and Object, Overloading, Access Privileges, Interface, Inner Class, Final and Static Modifiers, Packages, Inheritance, Overriding.
- 1.3. Handling Exceptions: Try, Catch, Finally, Throws, and Throw keywords, Creating Exception Class
- 1.4. Concurrency: Introduction, Thread States, Writing Multithreaded Programs, Thread Properties, Thread Synchronization, Thread Priorities
- 1.5. Working with Files: Byte Stream Classes, Character Stream Classes, Random Access File, Reading and Writing Objects.

#### **Unit 2: User Interface Components with Swing (10 Hrs.)**

- 2.1. Introduction: Concept of AWT, AWT vs Swing, Java Applets, Applet Life Cycle, Swing Class Hierarchy, Component and Containers
- 2.2. Layout Management: No Layout, Flow layout, Border Layout, Grid Layout, Gridbag Layout, Group Layout.
- 2.3. GUI Controls: Text Fields, Password Fields, Text Areas, Scroll Pane, Labels, Check Boxes, Radio Buttons, Borders, Combo Boxes, Sliders
- 2.4. Menu, Menu Item, Icons in Menu Items, Check Box and Radio Buttons in Menu Items, Pop-up Menus, Keyboard Mnemonics and Accelerators, Enabling and Disabling Menu Items, Toolbars, Tooltips
- 2.5. Option Dialogs, Creating Dialogs, File Choosers, Color Choosers, Internal Frames, Frames, Tables, Trees, and Tables.

**Unit 3: Event Handling (4 Hrs.)**

- 3.1. Event Handling Concept, Listener Interfaces, Using Action Commands, Adapter Classes
- 3.2. Handling Action Events, Key Events, Focus Events, Mouse Event, Window Event, Item Events

**Unit 4: Database Connectivity (4 Hrs.)**

- 4.1. JDBC Architecture, JDBC Driver Types, JDBC Configuration, Managing Connections, Statements, Result Set, SQL Exceptions
- 4.2. DDL and DML Operations using Java, Prepared Statements, Multiple Results, Scrollable Result Sets, Updateable Result Sets, Row Sets and Cached Row Sets, Transactions, SQL Escapes.

**Unit 5: Network Programming (5 Hrs.)**

- 5.1. Transmission control Protocol (TCP), User Datagram Protocol (UDP), Ports, IP Address Network Classes in JDK
- 5.2. Socket programming using TCP, Socket programming using UDP, Working with URL's, Working with URL Connection Class.
- 5.3. Java Mail API, Sending and Receiving Email

**Unit 6: GUI with JavaFX (3 Hrs.)**

- 6.1. Introduction, JavaFX vs Swing, JavaFX Layouts: FlowPane, BorderPane, Hbox, VBox, GridPane
- 6.2. JavaFX UI Controls: Label, TextField, Button, RadioButton, CheckBox, Hyperlink, Menu, Tooltip, FileChooser.

**Unit 7: Servlets and Java Server pages (8 Hrs.)**

- 7.1. Web Container, Introduction to Servlets, Life cycle of servlets, The servlet APIs, Writing Servlet Programs, Reading Form Parameters, Processing Forms, Handling HTTP Request and Response (GET / POST Request), Database Access with Servlets, Handling Cookies and Session.
- 7.2. Servlet vs JSP, JSP Access Model, JSP Syntax (Directions, Declarations, Expression, Scriptlets, Comments), JSP Implicit Objects, Object Scope, Processing Forms, Database Access with JSP.
- 7.3. Introduction to Java Web Frameworks

**Unit 8: RMI and CORBA (3 Hrs.)**

- 8.1 Introduction of RMI, Architecture of RMI, Creating and Executing RMI Applications
- 8.2 Introduction to CORBA, RMI vs CORBA, Architecture of CORBA, IDL, Simple CORBA Program.

**Laboratory Works:**

The laboratory work includes writing programs related to basic java programming concepts, Designing GUI, Event Handling, JDBC, Network Programming, Web Programming, and Distributed Programming. They also learn to develop web applications using Java Web Frameworks.

**Text Books:**

1. Cay S. Horstmann, Core Java Volume I--Fundamentals, Pearson, Eleventh Edition, 2018
2. Cay S. Horstmann, Core Java Volume II-Advance Features, Pearson, Eleventh Edition, 2019
3. Herbert Schildt, Java: The Complete Reference, McGraw-Hill Education, Eleventh Edition, 2018

**Reference Book:**

1. D.T. Editorial Services, Java 8 Programming Black Book, Dreamtech Press, 2015

## **Data Warehousing and Data Mining**

**Course Title:** Data Warehousing and Data Mining

**Course No:** CSC410

**Nature of the Course:** Theory + Lab

**Semester:** VII

**Full Marks:** 60 + 20 + 20

**Pass Marks:** 24 + 8 + 8

**Credit Hrs:** 3

### **Course Description:**

This course introduces advanced aspects of data warehousing and data mining, encompassing the principles, research results and commercial application of the current technologies.

### **Course Objective:**

The main objective of this course is to provide knowledge of different data mining techniques and data warehousing.

### **Course Contents:**

#### **Unit 1: Introduction to Data Warehousing (5 Hrs.)**

Lifecycle of data, Types of data, Data warehouse and data warehousing, Differences between operational database and data warehouse, A multidimensional data model, OLAP operation in multidimensional data model, Conceptual modeling of data warehouse, Architecture of data warehouse, Data warehouse implementation, Data marts, Components of data warehouse, Need for data warehousing, Trends in data warehousing

#### **Unit 2: Introduction to Data Mining (2 Hrs.)**

Motivation for data mining, Introduction to data mining system, Data mining functionalities, KDD, Data object and attribute types, Statistical description of data, Issues and Applications

#### **Unit 3: Data Preprocessing (3 Hrs.)**

Data cleaning, Data integration and transformation, Data reduction, Data discretization and Concept Hierarchy Generation, Data mining primitives

#### **Unit 4: Data Cube Technology (4 Hrs.)**

Efficient method for data cube computation, Cube materialization (Introduction to Full cube, Iceberg cube, Closed cube, Shell cube), General strategies for cube computation, Attribute oriented induction for data characterization, Mining class comparison, Discriminating between different classes

#### **Unit 5: Mining Frequent Patterns (6 Hrs.)**

Frequent patterns, Market basket analysis, Frequent itemsets, closed itemsets, association rules, Types of association rule (Single dimensional, multidimensional, multilevel, quantitative), Finding frequent itemset (Apriori algorithm, FP growth), Generating association rules from frequent itemset, Limitation and improving Apriori, From Association Mining to Correlation Analysis, Lift

**Unit 6: Classification and Prediction (10 Hrs.)**

Definition (Classification, Prediction), Learning and testing of classification, Classification by decision tree induction, ID3 as attribute selection algorithm, Bayesian classification, Laplace smoothing, Classification by backpropagation, Rule based classifier (Decision tree to rules, rule coverage and accuracy, efficient of rule simplification), Support vector machine, Evaluating accuracy (precision, recall, f-measure), Issues in classification, Overfitting and underfitting, K-fold cross validation, Comparing two classifier (McNemar's test)

**Unit 7: Cluster Analysis (8 Hrs.)**

Types of data in cluster analysis, Similarity and dissimilarity between objects, Clustering techniques: - Partitioning (k-means, k-means++, Mini-Batch k-means, k-medoids), Hierarchical (Agglomerative and Divisive), Density based (DBSCAN), Outlier analysis

**Unit 8: Graph Mining and Social Network Analysis (5 Hrs.)**

Graph mining, Why graph mining, Graph mining algorithm (Beam search, Inductive logic programming), Social network analysis, Link mining, Friends of friends, Degree assortativity, Signed network (Theory of structured balance, Theory of status, Conflict between the theory of balance and status), Trust in a network (Atomic propagation, Propagation of distrust, Iterative propagation), Predicting positive and negative links

**Unit 9: Mining Spatial, Multimedia, Text and Web Data (2 Hrs.)**

Spatial data mining, Spatial data cube, Mining spatial association, Multimedia data mining, Similarity search in multimedia data, Mining association in multimedia data, An introduction to text mining, natural language processing and information extraction, Web mining (Web content mining, Web structure mining, Web usage mining)

**Laboratory Works:**

The laboratory should contain all the features mentioned in a course, which should include data preprocessing and cleaning, implementing classification, clustering, association algorithms in any programming language, and data visualization through data mining tools.

**Text Book:**

1. Data Mining: Concepts and Techniques, 3rd ed. Jiawei Han, Micheline Kamber, and Jian Pei. Morgan Kaufmann Series in Data Management Systems Morgan Kaufmann Publishers, July 2011.

**Reference Books:**

1. Introduction to Data Mining, 2nd ed. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar. Pearson Publisher, 2019.
2. Mining of Massive Datasets by Jure Leskovec, Anand Rajaraman, Jeffrey D. Ullman, 2014.

## **Principles of Management**

**Course Title:** Principles of Management

**Course No:** MGT411

**Nature of the Course:** Theory

**Semester:** VII

**Full Marks:** 80 + 20

**Pass Marks:** 32 + 8

**Credit Hrs:** 3

### **Course Description:**

This course contains The Nature of Organizations, Introduction to Management, Evolution of Management Thought, Environmental Context of Management, Planning and Decision Making, Organizing Function, Leadership, Motivation, Communication, Control and Quality Management, Global Context of Management, Management Trends and Scenario in Nepal.

### **Course Objective:**

The basic objective of this course is to give a comprehensive knowledge to students about organization and help them understand the major functions, principles, and techniques of management. The course deals with basic functions like planning, organizing, leading, and controlling with special orientation to modern management practices which are essential to manage business successfully and other organizations.

### **Course Contents:**

#### **Unit 1: The Nature of Organizations (3 Hrs.)**

Concept of organization. Organizational goals – concept, purposes, and types. Features of effective organizational goals. Goal formulation – processes and approaches. Goal succession and displacement. Problems of goal formulation. Changing perspectives of organization.

#### **Unit 2: Introduction to Management (3 Hrs.)**

Definition, characteristics, and principles of management. Process and functions of management. Managerial hierarchy. Types of managers. Managerial skills and roles. Emerging challenges for management.

#### **Unit 3: Evolution of Management Thought (5 Hrs.)**

Introduction, contribution and limitation of Classical theory, Human relations and Behavioural science theories, System theory, Decision theory, Management science theory, and Contingency theory. Emerging management concepts: workforce diversity, outsourcing, knowledge management, learning organization.

#### **Unit 4: Environmental Context of Management (5 Hrs.)**

Concept of business environment. Types of business environment – internal and external. Basic components of economic, socio – cultural, political, and technological environments. Social responsibility of business – concept and approaches. Areas of social responsibility. Business ethics – meaning and significance. Emerging business environment in Nepal.

#### **Unit 5: Planning and Decision Making (5 Hrs.)**

Concept, types, hierarchy of planning. Process and importance of planning. Strategic planning. Environmental scanning – concept and methods. SWOT analysis. Formulation and

implementation of strategic plans. Quantitative tools for planning. Decision making – definition and approaches. Types of decisions. Decision making under conditions of certainty and uncertainty. Problem solving – concepts, types of problem. Problem solving strategies.

#### **Unit 6: Organizing Function (6 Hrs.)**

Concept and principles of organizing. Approaches to organizing – classical, behavioural, and contingency. Process of structuring an organization. Departmentalization – meaning and types. Delegation of authority – meaning, features, advantages, and barriers. Centralization and decentralization – meaning, advantages and disadvantages. Concept of organic and mechanistic views of organization. Types of modern organizational structures – matrix, team, and network.

#### **Unit 7: Leadership & Conflict (3 Hrs.)**

Concept and functions of leadership. Leadership styles. Approaches to leadership – trait, behavioral, and situational. Group formation. Types and characteristics of groups. Conflict – meaning and types. Managing conflicts in organization.

#### **Unit 8: Motivation (3 Hrs.)**

Concept. Theories of motivation – Need Hierarchy, and Motivation-Hygiene. Reward system to motivate performance. Motivation through employee participation – quality of work life, and self- managed teams.

#### **Unit 9: Communication (3 Hrs.)**

Concept, structure, and process. Types of communication – formal and informal. Interpersonal and nonverbal communication. Barriers to effective communication. Enhancing effective communication.

#### **Unit 10: Control and Quality Management (3 Hrs.)**

Concept, process, and types of control systems. Characteristics of effective control system. Quality control systems – concept of quality. Total Quality Management (TQM) – concept and tools. Deming management – principles and techniques.

#### **Unit 11: Global Context of Management (3 Hrs.)**

Concept of globalization. Methods of globalization. Effects of globalization. Multinational companies – meaning, types, advantages, and disadvantages.

#### **Unit 12: Management Trends and Scenario in Nepal (3 Hrs.)**

Growth of business sector in Nepal. Major industries in Nepal – manufacturing, export – oriented, import-substitution, and service sector. Existing management practices and business culture. Major problems of businesses in Nepal.

#### **Recommended Books:**

1. Griffin, Ricky W., **Management**, AITBS Publishers and Distributors, Delhi.
2. Hitt, Michael A., J. Black, Stewart, and Porter, Lyman W., **Management**, Pearson, India.
3. Robbins, Stephen P., and Coulter, Mary, **Management**, Prentice-Hall of India, New Delhi.

## **Software Project Management**

**Course Title:** Software Project Management

**Course No:** CSC415

**Nature of the Course:** Theory + Lab

**Semester:** VII

**Full Marks:** 60+20+20

**Pass Marks:** 24+8 + 8

**Credit Hrs:** 3

### **Course Description:**

This course familiarizes students with different concepts of software project management mainly focusing on project analysis, scheduling, resource allocation, risk analysis, monitoring, control and software configuration management.

### **Course Objectives:**

The main objective of this course is to provide knowledge of different concepts of software project management so that students will be able to understand and handle various projects including very high risky and innovative projects using different project management skills.

### **Course Contents:**

#### **Unit 1: Introduction to Software Project Management (5 Hrs.)**

Software engineering problem and software product, software product attributes, Definition of a Software Project (SP), SP Vs. other types of projects activities covered by SPM, categorizing SPs, Project management cycle, SPM framework, types of project plan.

#### **Unit 2: Project Analysis (8 Hrs.)**

Introduction, strategic assessment, technical assessment, economic analysis: Present worth, future worth, annual worth, internal rate of return (IRR) method, benefit-cost ratio analysis, including uniform gradient cash flow and comparison of mutually exclusive alternatives.

#### **Unit 3: Activity Planning and Scheduling (7 Hrs.)**

Objectives of activity planning, Work breakdown structure, Bar chart, Network planning model: Critical path method (CPM), Program evaluation and review technique (PERT), Precedence diagramming method (PDM), Shortening project duration, Identifying critical activities.

#### **Unit 4: Risk Management (4 Hrs.)**

Introduction, nature and identification of risk, risk analysis, evaluation of risk to the schedule using Z-values.

#### **Unit 5: Resource Allocation (4 Hrs.)**

Identifying resource requirements, resource allocation, resource smoothening and resource balancing.

#### **Unit 6: Monitoring and Control (4 Hrs.)**

Introduction, collecting data, visualizing progress, cost monitoring, earned value analysis, project control.



**Unit 7: Managing Contracts and people (5 Hrs.)**

Introduction, types of contract, stages in contract, placement, typical terms of a contract, contract management, acceptance, Managing people and organizing terms: Introduction, understanding behavior, organizational behavior: a back ground, selecting the right person for the job, instruction in the best methods, motivation, working in groups, becoming a team, decision making, leadership, organizational structures, conclusion, further exercises.

**Unit 8: Software quality assurance and testing (5 Hrs.)**

Testing principles and objectives, test plan, types and levels of testing, test strategies, program verification and validation, software quality, SEI-CMM, SQA activities, QA organization structure, SQA plan.

**Unit 9: Software Configuration Management (3 Hrs.)**

Introduction, need, basic configuration, management function, baseline, configuration management responsibilities.

**Laboratory / Project Work:**

Students should prepare a project report using different concepts of software project management. The project can be done in groups with at most four members in each group. Each group can select a case study and apply the concepts of software project management focusing on project analysis, scheduling, risk analysis, resource allocation, testing.

**Text Book:**

1. Software Project Management by Bob Hughes and Mike Cotterell, Latest Publication

**Reference Books:**

1. "Introduction to Software Project Management & Quality Assurance", Darrel Ince, I. Sharp, M. Woodman, Tata McGraw Hill
2. "Software Project Management: A Unified Framework", Walker Royce, Addison-Wesley, An Imprint of Pearson Education
3. "Managing the Software Process", Watts S. Humphrey, Addison-Wesley, An Imprint of Pearson Education